ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 4

NATURAL SCIENCES AND TECHNOLOGY

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 50

Marking guideline for the Academic Central Grade 4 Natural Sciences and Technology paper 1 · practice a.

Written by Academic Central for practice, following the CAPS curriculum for Grade 4. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words if the science is correct. Do not take marks off for spelling if the meaning is clear.

QUESTION 1

1	1.1 B ✓ 1.2 A ✓ 1.3 B ✓ 1.4 A ✓ 1.5 B ✓ 1.6 C ✓ 1.7 C ✓ 1.8 B ✓	(8)
		[8]

QUESTION 2

2	2.1 TRUE ✓ 2.2 FALSE ✓ 2.3 TRUE ✓ 2.4 TRUE ✓ 2.5 FALSE ✓ 2.6 TRUE ✓	(6)
		[6]

QUESTION 3

3.1	The Sun ✓	(1)
3.2	Grass ✓	(1)
3.3	It makes its own food, using energy from sunlight ✓	(1)
3.4	Rabbit ✓	(1)
3.5	Jackal ✓	(1)
3.6	The direction in which the food (energy) moves / "is eaten by" ✓	(1)
3.7	The jackals would go hungry, die or move away ✓ because they would have no food ✓	(2)
3.8	They eat both plants and animals ✓. Any correct example, e.g. pap with chicken, or rice with meat and vegetables ✓	(2)
3.9	Any TWO: to move, walk and play ✓; to grow ✓; to keep warm; to stay healthy; to breathe and keep the heart beating; to think and learn	(2)
		[12]

QUESTION 4

4.1	Any TWO: mains electricity (Eskom) ✓; batteries ✓; gas; paraffin; wood or coal; candles; the Sun (solar panels)	(2)
4.2	Electrical energy (electricity) ✓	(1)
4.3.1	Wind it up (twist it) around the axle ✓	(1)
4.3.2	Energy from Thandi's hand (muscles) when she winds up the elastic band ✓	(1)
4.3.3	Movement of the car ✓	(1)
4.3.4	Any ONE: wind the elastic band more times ✓; use a thicker or extra elastic band; let it run on a smooth, flat floor; make the car lighter	(1)
4.4	Mains electricity is very dangerous / strong ✓. It can give you an electric shock, burn you or kill you ✓	(2)

4.5 Any ONE: switch off lights and appliances you are not using ✓; boil only the water you need; unplug chargers; use the sun to dry clothes (1)

[10]

QUESTION 5

5.1 Day ✓ (1)

5.2 Night ✓ (1)

5.3 The Earth spins (turns) on its axis ✓. The side facing the Sun has day and the side turned away from the Sun has night ✓ (2)

5.4 Round, like a ball (a sphere) ✓ (1)

5.5 Any TWO: gives light ✓; gives heat (warmth) ✓; plants need sunlight to make food; it helps make rain (water cycle) (2)

5.6 It can damage your eyes or make you blind ✓ (1)

5.7 About 28 days (about a month) ✓ (accept 27 to 29 days) (1)

5.8 The phases of the Moon ✓ (1)

5.9 Any TWO: the Earth has air but the Moon has no air ✓; the Earth has water and the Moon has no liquid water ✓; there is life on Earth but not on the Moon; the Moon is smaller; the Moon has many craters; the Moon moves around the Earth (2)

5.10.1 Any ONE: there is no air to breathe on the Moon ✓; to protect them from the very hot and very cold temperatures (1)

5.10.2 Burning fuel pushes hot gases out downwards, and this pushes the rocket up ✓ (1)

[14]

TOTAL: 50

END OF MARKING GUIDELINE